

Reconocimiento de red en entorno Docker

Análisis de tráfico, descubrimiento de hosts y escaneo de puertos con Nmap y tcpdump

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Propósito: Demostración práctica de reconocimiento de red entre contenedores Docker

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Se inicia el entorno Docker ejecutando docker up para levantar los contenedores definidos en el compose. A continuación, se utiliza docker ps para verificar que los contenedores están en ejecución, mostrando su ID, estado y puertos expuestos.

```
May 4 16:30
Terminal
xnickh@xnickh-VMware-Virtual-Platform:~/Desktop/Reconocimiento-activo$ docker-compose -f docker-compose.yml up -d
Creating network "reconocimiento-activo_lab_network" with driver "bridge"
Creating attacker      ... done
Creating target-udp    ... done
Creating target-web    ... done
Creating target-icmp-only ... done
Creating target-multi  ... done
xnickh@xnickh-VMware-Virtual-Platform:~/Desktop/Reconocimiento-activo$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS
5564ba8ee4bf   alpine:latest  "sh -c ' apk add --n..." 4 seconds ago  Up 3 seconds  0.0.0.0:
2222->22/tcp, [::]:2222->22/tcp, 0.0.0.0:8081->8080/tcp, [::]:8081->8080/tcp  target-multi
89f06754aff3   alpine:latest  "sh -c 'tail -f /dev..." 4 seconds ago  Up 3 seconds  target-icmp-only
9ebd31bacff1   nginx:alpine   "/docker-entrypoint..." 4 seconds ago  Up 3 seconds  0.0.0.0:
8080->80/tcp, [::]:8080->80/tcp  target-web
446ab88def22   kalilinux/kali-rolling:latest "sleep infinity"         4 seconds ago  Up 3 seconds  attacker
bdd080cff5c9   alpine:latest  "sh -c ' apk add --n..." 4 seconds ago  Up Less than a second  target-udp
xnickh@xnickh-VMware-Virtual-Platform:~/Desktop/Reconocimiento-activo$
```

Se ejecuta el contenedor que actuará como atacante. Esto se realiza con docker exec -it <nombre_contenedor> /bin/bash, accediendo así a una terminal interactiva dentro del entorno controlado.

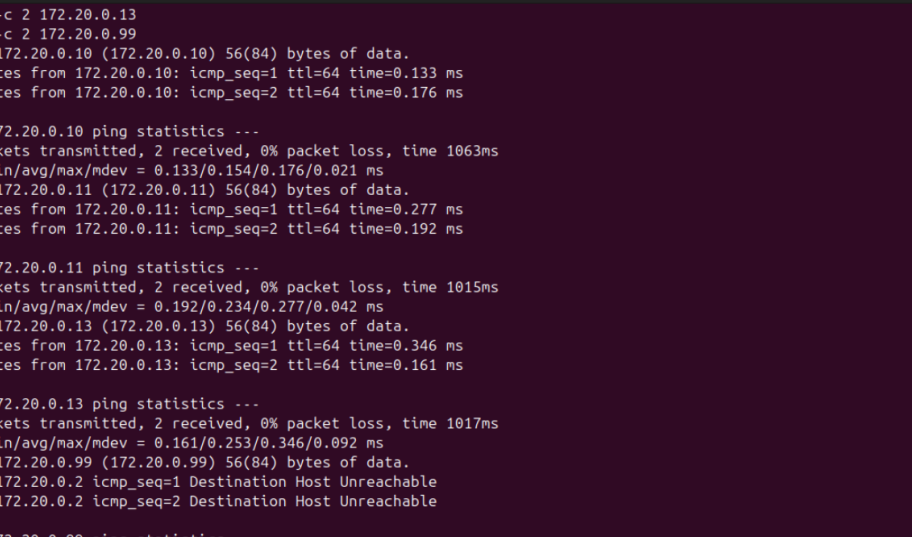
```
May 4 16:33
root@446ab88def22: ~
xnickh@xnickh-VMware-Virtual-Platform:~/Desktop/Reconocimiento-activo$ docker exec -it attacker bash
(root@446ab88def22)-[~]
# ping -c 2 172.20.0.10
ping -c 2 172.20.0.11
ping -c 2 172.20.0.13
ping -c 2 172.20.0.99
PING 172.20.0.10 (172.20.0.10) 56(84) bytes of data.
64 bytes from 172.20.0.10: icmp_seq=1 ttl=64 time=0.133 ms
64 bytes from 172.20.0.10: icmp_seq=2 ttl=64 time=0.176 ms

--- 172.20.0.10 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1063ms
rtt min/avg/max/mdev = 0.133/0.154/0.176/0.021 ms
PING 172.20.0.11 (172.20.0.11) 56(84) bytes of data.
64 bytes from 172.20.0.11: icmp_seq=1 ttl=64 time=0.277 ms
64 bytes from 172.20.0.11: icmp_seq=2 ttl=64 time=0.192 ms

--- 172.20.0.11 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1015ms
rtt min/avg/max/mdev = 0.192/0.234/0.277/0.042 ms
PING 172.20.0.13 (172.20.0.13) 56(84) bytes of data.
64 bytes from 172.20.0.13: icmp_seq=1 ttl=64 time=0.346 ms
64 bytes from 172.20.0.13: icmp_seq=2 ttl=64 time=0.161 ms

--- 172.20.0.13 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1017ms
rtt min/avg/max/mdev = 0.161/0.253/0.346/0.092 ms
PING 172.20.0.99 (172.20.0.99) 56(84) bytes of data.
From 172.20.0.2 icmp_seq=1 Destination Host Unreachable
From 172.20.0.2 icmp_seq=2 Destination Host Unreachable

--- 172.20.0.99 ping statistics ---
2 packets transmitted, 0 received, 100% packet loss, time 1016ms
```



```

root@446ab88def22: ~/src
ping -c 2 172.20.0.13
ping -c 2 172.20.0.99
PING 172.20.0.10 (172.20.0.10) 56(84) bytes of data.
64 bytes from 172.20.0.10: icmp_seq=1 ttl=64 time=0.133 ms
64 bytes from 172.20.0.10: icmp_seq=2 ttl=64 time=0.176 ms

--- 172.20.0.10 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1063ms
rtt min/avg/max/mdev = 0.133/0.154/0.176/0.021 ms
PING 172.20.0.11 (172.20.0.11) 56(84) bytes of data.
64 bytes from 172.20.0.11: icmp_seq=1 ttl=64 time=0.277 ms
64 bytes from 172.20.0.11: icmp_seq=2 ttl=64 time=0.192 ms

--- 172.20.0.11 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1015ms
rtt min/avg/max/mdev = 0.192/0.234/0.277/0.042 ms
PING 172.20.0.13 (172.20.0.13) 56(84) bytes of data.
64 bytes from 172.20.0.13: icmp_seq=1 ttl=64 time=0.346 ms
64 bytes from 172.20.0.13: icmp_seq=2 ttl=64 time=0.161 ms

--- 172.20.0.13 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1017ms
rtt min/avg/max/mdev = 0.161/0.253/0.346/0.092 ms
PING 172.20.0.99 (172.20.0.99) 56(84) bytes of data.
From 172.20.0.2 icmp_seq=1 Destination Host Unreachable
From 172.20.0.2 icmp_seq=2 Destination Host Unreachable

--- 172.20.0.99 ping statistics ---
2 packets transmitted, 0 received, +2 errors, 100% packet loss, time 1016ms
pipe 2

```

May 4 16:36

root@446ab88def22: ~/src

```
(root@446ab88def22): ~/src
# python3 host_discovery.py
```

HOST DISCOVERY TOOL - Práctica 2 Reconocimiento Activo
Implementación con Scapy

```

$
PRUEBA 1: Escaneo a host activo individual (172.20.0.10)
$
=====
INICIANDO DESCUBRIMIENTO DE HOSTS
Rango: 172.20.0.10
Protocolos: ['ICMP', 'TCP', 'UDP']
Puerto: 80
=====

[-] Probando con protocolo: ICMP
-----
[*] Construyendo 1 paquete(s) para ICMP
[*] Enviando 1 paquete(s) con timeout 2s
[+] ICMP: 1 respuesta(s) recibida(s)
    ✓ 172.20.0.10 → Respuesta ICMP (type 14)

[-] Probando con protocolo: TCP
-----
[*] Construyendo 1 paquete(s) para TCP
[+] Enviando 1 paquete(s) con timeout 2s
[+] TCP: 1 respuesta(s) recibida(s)
    ✓ 172.20.0.10 → Respuesta TCP (type 0)

```

La salida del script muestra las direcciones IP que responden activamente. En este caso, se identifica la IP del contenedor víctima y posiblemente la del gateway de la red Docker.

```
May 4 16:36
root@446ab88def22: ~/src

[*] Construyendo 1 paquete(s) para ICMP
[*] Enviando 1 paquete(s) con timeout 2s
[+] ICMP: 1 respuesta(s) recibida(s)
    ✓ 172.20.0.10 → Respuesta ICMP (type 14)

[->] Probando con protocolo: TCP
-----
[*] Construyendo 1 paquete(s) para TCP
[*] Enviando 1 paquete(s) con timeout 2s
[+] TCP: 1 respuesta(s) recibida(s)
    ✓ 172.20.0.10 → Respuesta TCP (flags R)

[->] Probando con protocolo: UDP
-----
[*] Construyendo 1 paquete(s) para UDP
[*] Enviando 1 paquete(s) con timeout 4s
[+] UDP: 1 respuesta(s) recibida(s)
    ✓ 172.20.0.10 → Respondió

=====
REPORTE FINAL DE DESCUBRIMIENTO
=====
Rango escaneado: 172.20.0.10
Tiempo total: 1.63 segundos
Hosts activos encontrados: 1

Lista de hosts activos:
1. 172.20.0.10

Detalles por host:
```

```
May 4 16:37
root@446ab88def22: ~/src

Detalles por host:
  172.20.0.10
  Protocolos que respondieron: UDP, TCP, ICMP
    ├── ICMP type 14
    ├── TCP flags R
    └── Respuesta recibida

=====
$
PRUEBA 2: Escaneo a IP sin host activo (172.20.0.99)
$
=====
INICIANDO DESCUBRIMIENTO DE HOSTS
Rango: 172.20.0.99
Protocolos: ['ICMP', 'TCP']
Puerto: 80
=====

[->] Probando con protocolo: ICMP
-----
[*] Construyendo 1 paquete(s) para ICMP
[*] Enviando 1 paquete(s) con timeout 2s
WARNING: MAC address to reach destination not found. Using broadcast.
[-] ICMP: No se recibieron respuestas
```

```
May 4 16:38
root@446ab88def22: ~/src

[-] Probando con protocolo: TCP
-----
[*] Construyendo 1 paquete(s) para TCP
[*] Enviando 1 paquete(s) con timeout 2s
WARNING: MAC address to reach destination not found. Using broadcast.
[-] TCP: No se recibieron respuestas

=====
REPORTE FINAL DE DESCUBRIMIENTO
=====
Rango escaneado: 172.20.0.99
Tiempo total: 9.12 segundos
Hosts activos encontrados: 0

[-] No se encontraron hosts activos en el rango especificado

=====
PRUEBA 3: Escaneo de rango completo (172.20.0.0/24)
=====
INICIANDO DESCUBRIMIENTO DE HOSTS
Rango: 172.20.0.0/24
Protocolos: ['ICMP', 'TCP']
Puerto: 80
=====

[-] Probando con protocolo: ICMP
```

```
May 4 16:39
root@446ab88def22: ~/src

=====

[-] Probando con protocolo: ICMP
-----
[*] Construyendo 1 paquete(s) para ICMP
[*] Enviando 1 paquete(s) con timeout 2s
WARNING: MAC address to reach destination not found. Using broadcast.
[-] ICMP: No se recibieron respuestas

[-] Probando con protocolo: TCP
-----
[*] Construyendo 1 paquete(s) para TCP
[*] Enviando 1 paquete(s) con timeout 2s
WARNING: MAC address to reach destination not found. Using broadcast.
[-] TCP: No se recibieron respuestas

=====
REPORTE FINAL DE DESCUBRIMIENTO
=====
Rango escaneado: 172.20.0.0/24
Tiempo total: 9.13 segundos
Hosts activos encontrados: 0

[-] No se encontraron hosts activos en el rango especificado

=====
PRUEBA 4: Escaneo rápido con timeout reducido
=====
```


Como Nmap no está disponible por defecto, se instala dentro del contenedor atacante usando `apt update && apt install nmap -y`. Esto permite realizar escaneos de puertos directamente desde el entorno aislado. Se ejecuta `nmap -sS -p- -Pn <IP_victima>` para realizar un escaneo sigiloso de todos los puertos. La opción `-Pn` evita el ping inicial, útil si la víctima bloquea ICMP.

```
root@446ab88def22: ~/src
# # Instalar nmap
apt-get install -y nmap

# Escaneo por defecto a target-web
nmap 172.20.0.10

# Escaneo a target-multi
nmap 172.20.0.11

# Escaneo a target-icmp-only
nmap 172.20.0.13

# Puerto cerrado
nmap -p 22 172.20.0.10
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Solving dependencies... Done
The following additional packages will be installed:
  liblinear4 liblua5.4-0 libssh2-1t64 nmap-common
Suggested packages:
  liblinear-tools liblinear-dev ncat ndiff zenmap
The following NEW packages will be installed:
  liblinear4 liblua5.4-0 libssh2-1t64 nmap nmap-common
0 upgraded, 5 newly installed, 0 to remove and 17 not upgraded.
Need to get 7120 kB of archives.
After this operation, 29.9 MB of additional disk space will be used.
Get:1 http://http.kali.org/kali kali-rolling/main amd64 liblinear4 amd64 2.3.0+dfsg-5+b3 [42.9 kB]
Get:2 http://http.kali.org/kali kali-rolling/main amd64 liblua5.4-0 amd64 5.4.8-1+b2 [148 kB]
Get:3 http://mirror.raiolanetworks.com/kali kali-rolling/main amd64 libssh2-1t64 amd64 1.11.1-2 [244 kB]
Get:5 http://http.kali.org/kali kali-rolling/non-free amd64 nmap amd64 7.99+dfsg-1kali1 [1987 kB]
Get:4 http://http.kali.org/kali kali-rolling/non-free amd64 nmap-common all 7.99+dfsg-1kali1 [4608 kB]
```

```
Unpacking nmap-common (7.99+dfsg-1kali1) ...
Selecting previously unselected package nmap.
Preparing to unpack .../nmap_7.99+dfsg-1kali1_amd64.deb ...
Unpacking nmap (7.99+dfsg-1kali1) ...
Setting up liblinear4:amd64 (2.3.0+dfsg-5+b3) ...
Setting up nmap-common (7.99+dfsg-1kali1) ...
Setting up liblua5.4-0:amd64 (5.4.8-1+b2) ...
Setting up libssh2-1t64:amd64 (1.11.1-2) ...
Setting up nmap (7.99+dfsg-1kali1) ...
Processing triggers for libc-bin (2.42-13) ...
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-04 14:41 +0000
Nmap scan report for target-web.reconocimiento-activo_lab_network (172.20.0.10)
Host is up (0.0000030s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE
80/tcp    open  http
MAC Address: 02:2C:D0:A7:E9:DA (Unknown)

Nmap done: 1 IP address (1 host up) scanned in 0.15 seconds
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-04 14:41 +0000
Nmap scan report for target-multi.reconocimiento-activo_lab_network (172.20.0.11)
Host is up (0.0000030s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
MAC Address: 32:17:A9:CC:69:D8 (Unknown)

Nmap done: 1 IP address (1 host up) scanned in 0.14 seconds
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-04 14:41 +0000
Nmap scan report for target-icmp-only.reconocimiento-activo_lab_network (172.20.0.13)
Host is up (0.0000040s latency).
All 1000 scanned ports on target-icmp-only.reconocimiento-activo_lab_network (172.20.0.13) are in ignored states.
```

```
May 4 16:41
root@446ab88def22: ~/src
MAC Address: 02:2C:D0:A7:E9:DA (Unknown)

Nmap done: 1 IP address (1 host up) scanned in 0.15 seconds
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-04 14:41 +0000
Nmap scan report for target-multi.reconocimiento-activo_lab_network (172.20.0.11)
Host is up (0.0000030s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
MAC Address: 32:17:A9:CC:69:D8 (Unknown)

Nmap done: 1 IP address (1 host up) scanned in 0.14 seconds
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-04 14:41 +0000
Nmap scan report for target-icmp-only.reconocimiento-activo_lab_network (172.20.0.13)
Host is up (0.0000040s latency).
All 1000 scanned ports on target-icmp-only.reconocimiento-activo_lab_network (172.20.0.13) are in ignored states.
Not shown: 1000 closed tcp ports (reset)
MAC Address: 22:41:02:C5:8E:E2 (Unknown)

Nmap done: 1 IP address (1 host up) scanned in 0.14 seconds
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-04 14:41 +0000
Nmap scan report for target-web.reconocimiento-activo_lab_network (172.20.0.10)
Host is up (0.000039s latency).

PORT      STATE SERVICE
22/tcp    closed ssh
MAC Address: 02:2C:D0:A7:E9:DA (Unknown)

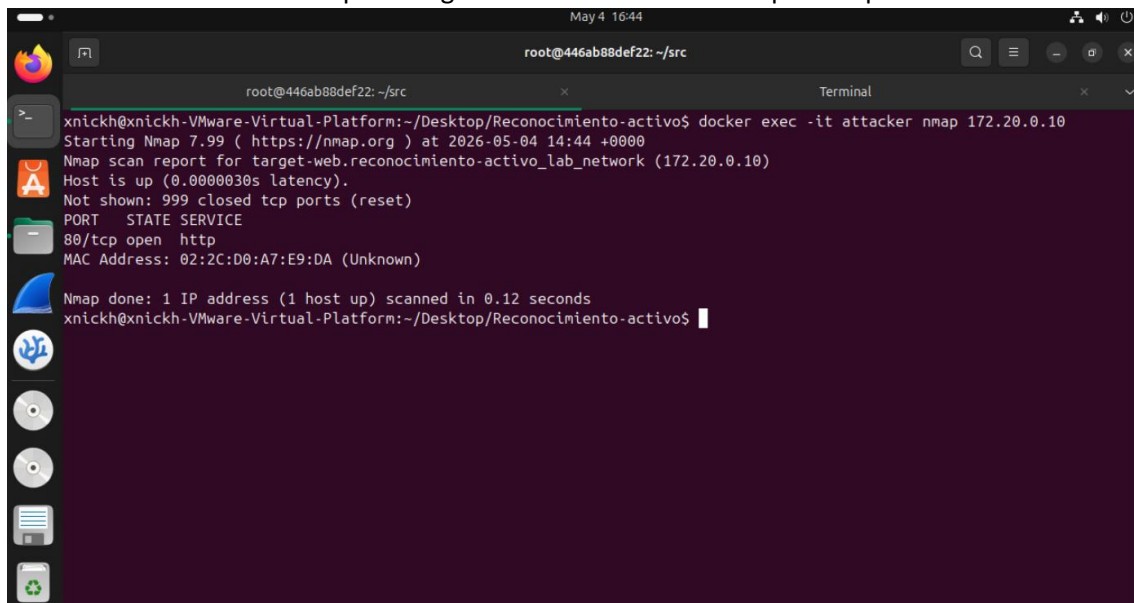
Nmap done: 1 IP address (1 host up) scanned in 0.11 seconds
(root@446ab88def22)~[/src]
```

En una segunda terminal, se inicia tcpdump -i eth0 -w captura.pcap para capturar todo el tráfico de red mientras se ejecuta el escaneo. Esto permitirá analizar después los paquetes enviados y recibidos.

```
May 4 16:44
Terminal
root@446ab88def22: ~/src
Terminal

xnckh@xnckh-VMware-Virtual-Platform:~/Desktop/Reconocimiento-activo$ cd ..
xnckh@xnckh-VMware-Virtual-Platform:~/Desktop$ ip a | grep br-
33: br-857c923fd419: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default
    inet 172.20.0.1/24 brd 172.20.0.255 scope global br-857c923fd419
35: veth24ed2e9@if2: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue master br-857c923fd419 state UP group default
36: vethb904bdd@if2: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue master br-857c923fd419 state UP group default
37: veth33f7c60@if2: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue master br-857c923fd419 state UP group default
38: veth0b55594@if2: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue master br-857c923fd419 state UP group default
xnckh@xnckh-VMware-Virtual-Platform:~/Desktop$ sudo tcpdump -i br-857c923fd419 -w captura_nmap.pcap
[sudo] password for xnckh:
tcpdump: listening on br-857c923fd419, link-type EN10MB (Ethernet), snapshot length 262144 bytes
```

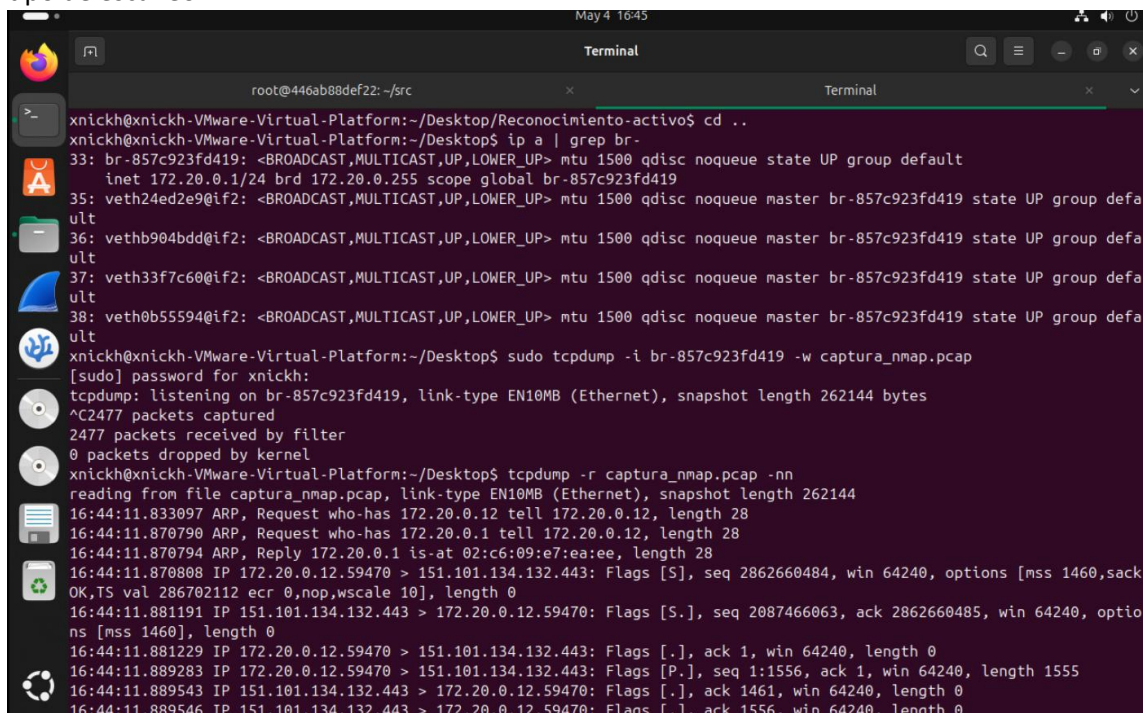

Mientras tcpdump está en ejecución, se lanza el escaneo de Nmap desde el contenedor atacante. Toda la actividad queda registrada en el archivo PCAP para su posterior análisis



```
root@446ab88def22: ~/src
xnickh@xnckh-VMware-Virtual-Platform: ~/Desktop/Reconocimiento-activo$ docker exec -it attacker nmap 172.20.0.10
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-04 14:44 +0000
Nmap scan report for target-web.reconocimiento-activo_lab_network (172.20.0.10)
Host is up (0.0000030s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE
80/tcp    open  http
MAC Address: 02:2C:D0:A7:E9:DA (Unknown)

Nmap done: 1 IP address (1 host up) scanned in 0.12 seconds
xnckh@xnckh-VMware-Virtual-Platform: ~/Desktop/Reconocimiento-activo$
```

Una vez terminado el escaneo, se detiene tcpdump y se visualiza la captura con tcpdump -r captura.pcap -n. Se observan paquetes TCP SYN, SYN-ACK, RST y posiblemente ICMP, según el tipo de escaneo.



```
root@446ab88def22: ~/src
xnickh@xnckh-VMware-Virtual-Platform: ~/Desktop/Reconocimiento-activo$ cd ..
xnickh@xnckh-VMware-Virtual-Platform: ~/Desktop$ ip a | grep br-
33: br-857c923fd419: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default
    inet 172.20.0.1/24 brd 172.20.0.255 scope global br-857c923fd419
35: veth24ed2e9@if2: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue master br-857c923fd419 state UP group default
    uld
36: vethb904bdd@if2: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue master br-857c923fd419 state UP group default
    uld
37: veth33f7c60@if2: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue master br-857c923fd419 state UP group default
    uld
38: veth0b55594@if2: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue master br-857c923fd419 state UP group default
    uld
xnickh@xnckh-VMware-Virtual-Platform: ~/Desktop$ sudo tcpdump -i br-857c923fd419 -w captura_nmap.pcap
[sudo] password for xnckh:
tcpdump: listening on br-857c923fd419, link-type EN10MB (Ethernet), snapshot length 262144 bytes
^C2477 packets captured
2477 packets received by filter
0 packets dropped by kernel
xnickh@xnckh-VMware-Virtual-Platform: ~/Desktop$ tcpdump -r captura_nmap.pcap -n
reading from file captura_nmap.pcap, link-type EN10MB (Ethernet), snapshot length 262144
16:44:11.833097 ARP, Request who-has 172.20.0.12 tell 172.20.0.12, length 28
16:44:11.870790 ARP, Request who-has 172.20.0.1 tell 172.20.0.12, length 28
16:44:11.870794 ARP, Reply 172.20.0.1 is-at 02:c6:09:e7:ea:ee, length 28
16:44:11.870808 IP 172.20.0.12.59470 > 151.101.134.132.443: Flags [S], seq 2862660484, win 64240, options [mss 1460,sack
OK,TS val 286702112 ecr 0,nop,wscale 10], length 0
16:44:11.881191 IP 151.101.134.132.443 > 172.20.0.12.59470: Flags [S.], seq 2087466063, ack 2862660485, win 64240, optio
ns [mss 1460], length 0
16:44:11.881229 IP 172.20.0.12.59470 > 151.101.134.132.443: Flags [.], ack 1, win 64240, length 0
16:44:11.889283 IP 172.20.0.12.59470 > 151.101.134.132.443: Flags [P.], seq 1:1556, ack 1, win 64240, length 1555
16:44:11.889543 IP 151.101.134.132.443 > 172.20.0.12.59470: Flags [.], ack 1461, win 64240, length 0
16:44:11.889546 IP 151.101.134.132.443 > 172.20.0.12.59470: Flags [F.], ack 1556, win 64240, length 0
```

```
May 4 16:46
Terminal
root@446ab88def22: ~/src
16:45:12.701377 IP 199.232.34.132.443 > 172.20.0.12.53840: Flags [.], seq 2660310:2661770, ack 2029, win 64240, length 1460
16:45:12.701465 IP 199.232.34.132.443 > 172.20.0.12.53840: Flags [P.], seq 2661770:2711409, ack 2029, win 64240, length 49639
16:45:12.701526 IP 199.232.34.132.443 > 172.20.0.12.53840: Flags [.], seq 2711409:2723089, ack 2029, win 64240, length 11680
16:45:12.701554 IP 199.232.34.132.443 > 172.20.0.12.53840: Flags [P.], seq 2723089:2724549, ack 2029, win 64240, length 1460
16:45:12.706856 IP 172.20.0.12.53840 > 199.232.34.132.443: Flags [.], ack 2724549, win 65535, length 0
16:45:12.707152 IP 199.232.34.132.443 > 172.20.0.12.53840: Flags [P.], seq 2724549:2788788, ack 2029, win 64240, length 64239
16:45:12.707204 IP 172.20.0.12.53840 > 199.232.34.132.443: Flags [.], ack 2788788, win 65535, length 0
16:45:12.707401 IP 199.232.34.132.443 > 172.20.0.12.53840: Flags [P.], seq 2788788:2853027, ack 2029, win 64240, length 64239
16:45:12.714528 IP 172.20.0.12.53840 > 199.232.34.132.443: Flags [.], ack 2853027, win 65535, length 0
16:45:12.715094 IP 199.232.34.132.443 > 172.20.0.12.53840: Flags [P.], seq 2853027:2915779, ack 2029, win 64240, length 62752
16:45:12.715136 IP 172.20.0.12.53840 > 199.232.34.132.443: Flags [.], ack 2915779, win 65535, length 0
16:45:12.835718 IP 172.20.0.12.53840 > 199.232.34.132.443: Flags [P.], seq 2029:2053, ack 2915779, win 65535, length 24
16:45:12.836016 IP 199.232.34.132.443 > 172.20.0.12.53840: Flags [.], ack 2053, win 64240, length 0
16:45:12.836556 IP 172.20.0.12.53840 > 199.232.34.132.443: Flags [F.], seq 2053, ack 2915779, win 65535, length 0
16:45:12.836758 IP 199.232.34.132.443 > 172.20.0.12.53840: Flags [.], ack 2054, win 64239, length 0
16:45:12.845153 IP 199.232.34.132.443 > 172.20.0.12.53840: Flags [FP.], seq 2915779:2915803, ack 2054, win 64239, length 24
16:45:12.845182 IP 172.20.0.12.53840 > 199.232.34.132.443: Flags [R], seq 55310757, win 0, length 0
xnckh@xnckh-VMware-Virtual-Platform:~/Desktop$ tcpdump -r captura_nmap.pcap -nn | wc -l
reading from file captura_nmap.pcap, link-type EN10MB (Ethernet), snapshot length 262144
2477
```

Se repite el procedimiento, pero esta vez cambiando el tipo de escaneo (por ejemplo, -sT para conexión completa) o modificando los puertos a escanear, con el fin de comparar el tráfico generado.

```
May 4 17:04
Terminal
root@446ab88def22: ~/src
xnckh@xnckh-VMware-Virtual-Platform:~/Desktop$ sudo tcpdump -i br-857c923fd419 -w captura_nmap_limpiar.pcap
[sudo] password for xnckh:
tcpdump: br-857c923fd419: No such device exists
(No such device exists)
xnckh@xnckh-VMware-Virtual-Platform:~/Desktop$ ip a | grep br-
94: br-bda953d62630: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default
    inet 172.20.0.1/24 brd 172.20.0.255 scope global br-bda953d62630
95: vethdf6f7a2@if2: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue master br-bda953d62630 state UP group default
96: vethc29aca6@if2: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue master br-bda953d62630 state UP group default
97: vethda96db0@if2: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue master br-bda953d62630 state UP group default
98: veth60b07f8@if2: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue master br-bda953d62630 state UP group default
xnckh@xnckh-VMware-Virtual-Platform:~/Desktop$ sudo tcpdump -i br-857c923fd419 -w captura_nmap_limpiar.pcap
tcpdump: br-857c923fd419: No such device exists
(No such device exists)
xnckh@xnckh-VMware-Virtual-Platform:~/Desktop$ sudo tcpdump -i br-bda953d62630 -w captura_nmap_limpiar.pcap
tcpdump: listening on br-bda953d62630, link-type EN10MB (Ethernet), snapshot length 262144 bytes
```

```
May 4 17:04
root@446ab88def22: ~/src
xnickh@xnickh-VMware-Virtual-Platform: ~/Desktop/Reconocimiento-activo$ docker exec -it attacker bash -c "nmap 172.20.0.10"
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-04 15:04 +0000
Nmap scan report for target-web.reconocimiento-activo_lab_network (172.20.0.10)
Host is up (0.0000030s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE
80/tcp    open  http
MAC Address: E2:42:74:F7:F0:01 (Unknown)
Nmap done: 1 IP address (1 host up) scanned in 0.13 seconds
xnickh@xnickh-VMware-Virtual-Platform: ~/Desktop/Reconocimiento-activo$
```

```
May 4 17:05
Terminal
root@446ab88def22: ~/src
17:04:50.561227 IP 172.20.0.10.4004 > 172.20.0.2.36556: Flags [R.], seq 0, ack 3112125456, win 0, length 0
17:04:50.561231 IP 172.20.0.2.36556 > 172.20.0.10.3918: Flags [S], seq 3112125455, win 1024, options [mss 1460], length 0
17:04:50.561233 IP 172.20.0.10.3918 > 172.20.0.2.36556: Flags [R.], seq 0, ack 3112125456, win 0, length 0
17:04:50.561236 IP 172.20.0.2.36556 > 172.20.0.10.3000: Flags [S], seq 3112125455, win 1024, options [mss 1460], length 0
17:04:50.561238 IP 172.20.0.10.3000 > 172.20.0.2.36556: Flags [R.], seq 0, ack 3112125456, win 0, length 0
17:04:50.561241 IP 172.20.0.2.36556 > 172.20.0.10.726: Flags [S], seq 3112125455, win 1024, options [mss 1460], length 0
17:04:50.561243 IP 172.20.0.10.726 > 172.20.0.2.36556: Flags [R.], seq 0, ack 3112125456, win 0, length 0
17:04:50.561246 IP 172.20.0.2.36556 > 172.20.0.10.9876: Flags [S], seq 3112125455, win 1024, options [mss 1460], length 0
17:04:50.561247 IP 172.20.0.10.9876 > 172.20.0.2.36556: Flags [R.], seq 0, ack 3112125456, win 0, length 0
17:04:50.561250 IP 172.20.0.2.36556 > 172.20.0.10.5801: Flags [S], seq 3112125455, win 1024, options [mss 1460], length 0
17:04:50.561252 IP 172.20.0.10.5801 > 172.20.0.2.36556: Flags [R.], seq 0, ack 3112125456, win 0, length 0
17:04:50.561255 IP 172.20.0.2.36556 > 172.20.0.10.5678: Flags [S], seq 3112125455, win 1024, options [mss 1460], length 0
17:04:50.561257 IP 172.20.0.10.5678 > 172.20.0.2.36556: Flags [R.], seq 0, ack 3112125456, win 0, length 0
17:04:50.561260 IP 172.20.0.2.36556 > 172.20.0.10.42: Flags [S], seq 3112125455, win 1024, options [mss 1460], length 0
17:04:50.561262 IP 172.20.0.10.42 > 172.20.0.2.36556: Flags [R.], seq 0, ack 3112125456, win 0, length 0
17:04:50.561265 IP 172.20.0.2.36556 > 172.20.0.10.8200: Flags [S], seq 3112125455, win 1024, options [mss 1460], length 0
17:04:50.561267 IP 172.20.0.10.8200 > 172.20.0.2.36556: Flags [R.], seq 0, ack 3112125456, win 0, length 0
17:04:55.913472 ARP, Request who-has 172.20.0.2 tell 172.20.0.10, length 28
17:04:55.913484 ARP, Reply 172.20.0.2 is-at 96:e4:b1:79:b7:00, length 28
xnickh@xnickh-VMware-Virtual-Platform: ~/Desktop$ tcpdump -r captura_nmap_limpiar.pcap -nn host 172.20.0.10 and host 172.20.0.2 | wc -l
reading from file captura_nmap_limpiar.pcap, link-type EN10MB (Ethernet), snapshot length 262144
2005
xnickh@xnickh-VMware-Virtual-Platform: ~/Desktop$
```

Se inicia nuevamente tcpdump y se ejecuta el segundo escaneo. Es importante limpiar las capturas anteriores o usar un nombre diferente para no sobrescribir los datos.

Al revisar la segunda captura, se pueden notar diferencias en los flags TCP, reintentos, respuestas ICMP tipo 3 (puerto inaccesible) o patrones de conexión más intrusivos según el método usado.

Finalmente, se han completado todas las fases del reconocimiento:

1. Verificación de conectividad.
2. Descubrimiento de hosts.
3. Escaneo de puertos con Nmap.
4. Captura y análisis de tráfico con tcpdump.

Los resultados permiten identificar servicios potencialmente vulnerables en la víctima.